



# ASSESSMENT COVERSHEET

## ENG80002 Engineering Master Major Thesis

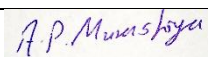
**Project Title:** Comparative Analysis of ANN and PID Controllers for VSC-HVDC Transmission Network

**Project ID:**

**Supervisor's Name:** Mehdi Korki

**Name of Industry** (for Industry sponsored project only):

### Student Details:

| Student ID | Given Name* | Family Name* | Major*     | Signature                                                                          | Date       |
|------------|-------------|--------------|------------|------------------------------------------------------------------------------------|------------|
| 104528108  | Aniruddh    | Murasiya     | Electrical |  | 27/05/2025 |

\* As it appears in the Canvas.

\* Major categorises into *Civil, Structural, Mechanical, Advanced Manufacturing, Electrical and Electronic and Network Systems and Telecommunications*

### DECLARATION

I declare that ( the first four boxes must be completed for the assignment/report to be accepted):

- ☒ This assignment/report contains no material that has previously been submitted for assessment at this or any other university. It contains no materials previously published or written by another person except where due reference has been made. Where a report has been professionally edited, the student must include separate acknowledgement that such "editing has been and the editing has addressed only the style and/or grammar of the report and not its substantive content."
- ☒ This is an original piece of work and no part has been completed by any other student than those signed below.
- ☒ I have read and understood the avoiding plagiarism guidelines at <http://www.swinburne.edu.au/ltas/plagiarism/students.htm> and no part of this work has been copied or paraphrased **from any other source** except where this has been clearly acknowledged in the body of the assignment and included in the reference list.
- ☒ I have retained a copy of this assignment/report in the event of it becoming lost or damaged.
- ☒ (optional) I agree to a copy of the assignment/report being retained as an exemplar for future students (subject to identifying details being removed).

**Date of Submission:** 27/05/2025

# Comparative Analysis of ANN and PID Controllers for VSC-HVDC Transmission Network

Supervisor

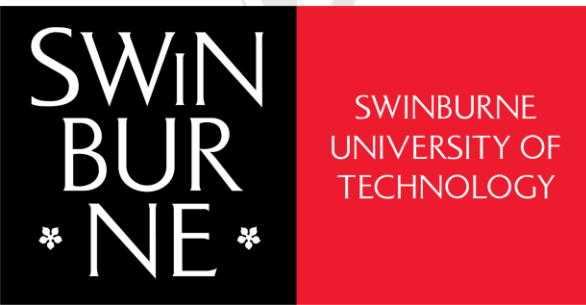
Dr. Mehdi Korki

Student

Aniruddh Murasiya

Student Id

104528108



ENG80002 Engineering Master Major Thesis

# ● **High-voltage Direct current** is vital for long-distance, **stable power** transfer

## Introduction

Type of stability in HVDC systems:

- Voltage stability - Maintaining DC voltage within an acceptable limit
- Current stability - Maintaining constant current flow
- Control stability – Provides stable reference tracking without Oscillations

---

Aim & Objective

Methods

Analysis & Results

Conclusion

# PID – Proportional Integral Derivative control

Maintain the **stability** of an HVDC transmission system by continuously adjusting control inputs to minimise error

**Proportional  
(P)**

React to  
current  
error

**Integral  
(I)**

React to  
accumulated past  
error

**Derivative  
(D)**

Predicts future  
error based on the  
rate of change



# Why Artificial Neural Network (ANN) over PID?

- Nonlinear Control Capability
- Adaptive learning
- No need for an Explicit Mathematical model
- Better fault tolerance
- Predictive capabilities
- Multi-variable control support
- Reduced Manual tuning

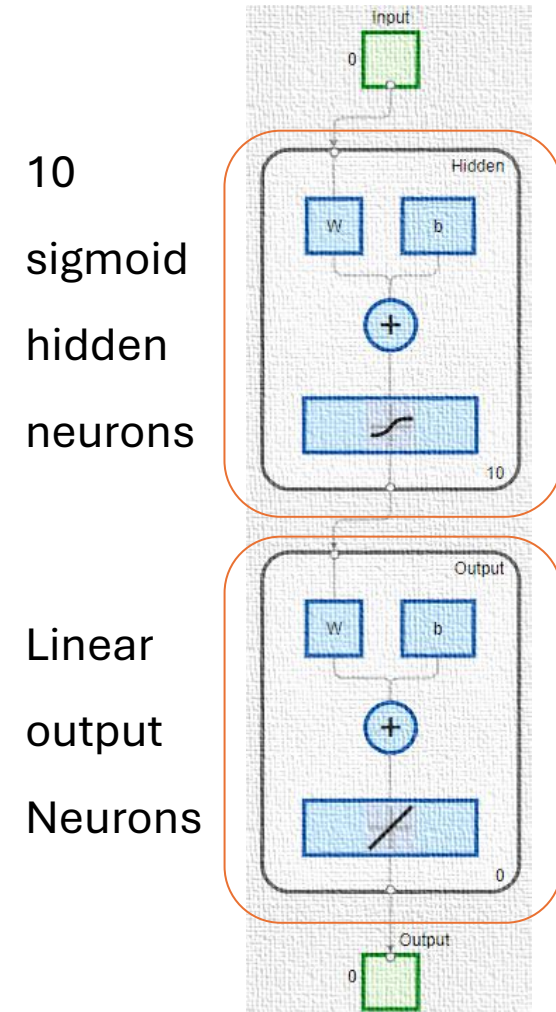


Fig.1 ANN network topology, Image taken from MATLAB.

# Enhance HVDC control performance

Aim & Objective

- Compare Active/Reactive power control of ANN and PID
- ANN outperform PID
- Achieve better stability using ANN

## Gap and significance

- Use of PID as a learning baseline
- Use of identical test conditions
- Used Renewable energy on the generating side



# Key Objectives

- a. Model a two-terminal VSC-HVDC transmission system in MATLAB/Simulink with a PID control strategy
- b. Generate training data with the PID controller
- c. Train ANN on PID response data
- d. Conduct testing under disturbance with ANN controller
- e. Compare the performance of ANN and PID controllers
- f. Evaluate the potential of ANN as a replacement of PID

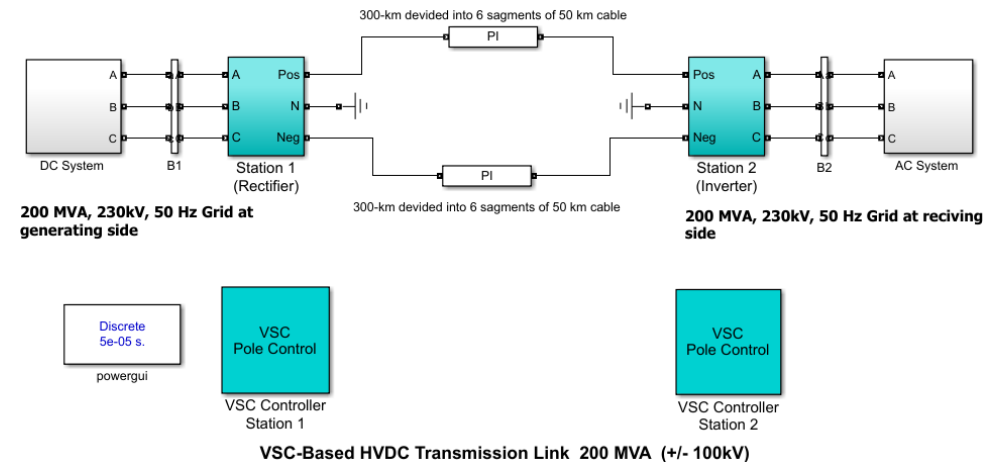


Fig. 3 VSC-HVDC transmission line, 200MAV (+/- 100 kV), Simulated in Simscap Electrical/Simulink.

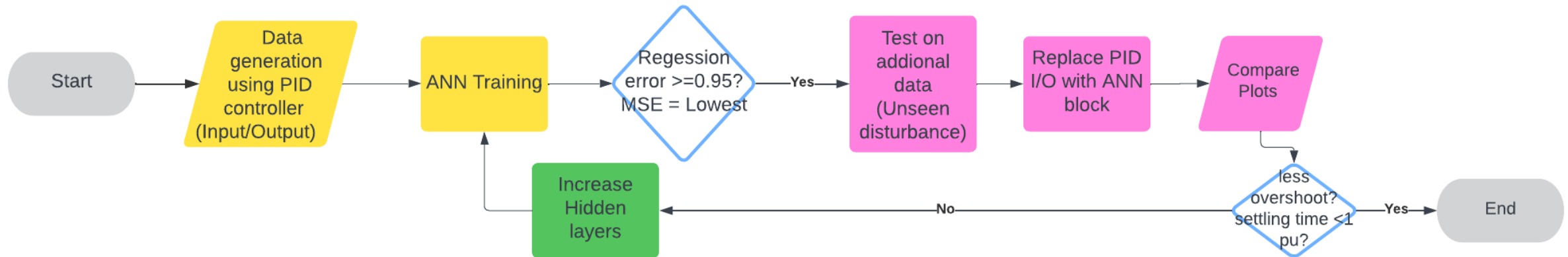
# Novelty Analysis

- ANN Trained on PID Response Data
- Hybrid Controller Architecture
- Improved Fault Resilience
- Knowledge Transfer through Supervised Learning



# Flowchart

## Methods



Introduction

Aim & Objective

Analysis & Results

Conclusion

# Voltage source Control (VSC)

# Outer Active/Reactive Power and Voltage loop

## Input signals

- Actual Active/Reactive power and Actual voltage ( $P_{meas}$ ,  $Q_{meas}$ ,  $V_{meas}$  and  $V_{dc\_meas}$ )
- Reference Active/Reactive power and voltage ( $P_{ref}$ ,  $Q_{ref}$ ,  $U_{f\_meas}$ )

## Output Signals

- d-axis and q-axis Reference current ( $I_{d\_ref}$  and  $I_{q\_ref}$ )

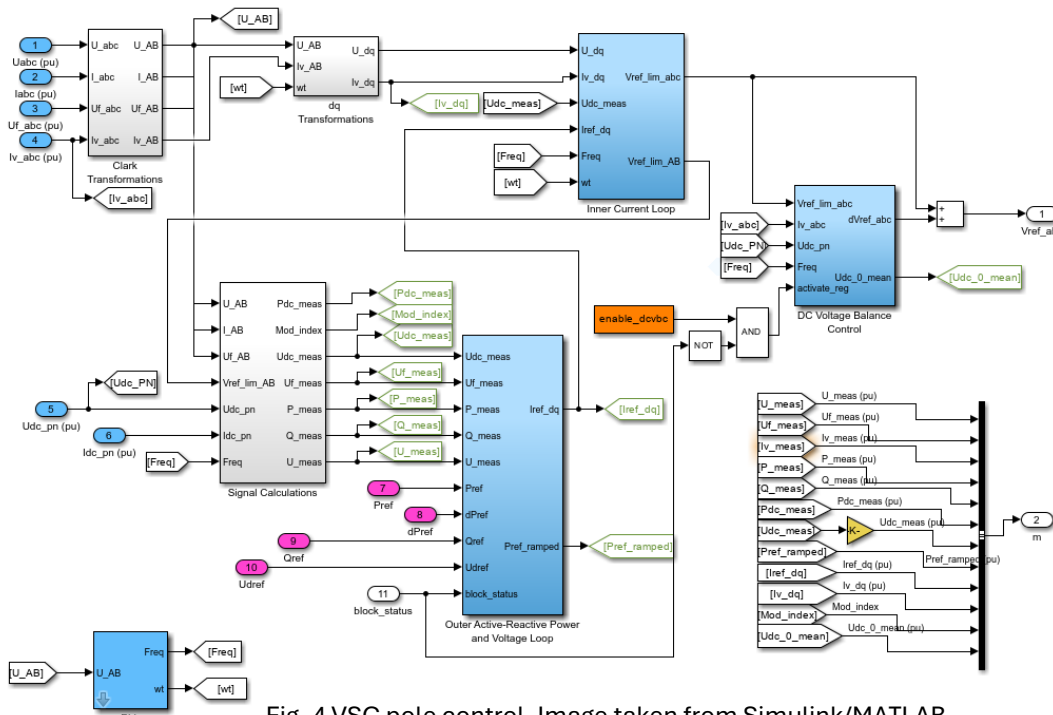


Fig. 4 VSC pole control, Image taken from Simulink/MATLAB.

# Disturbances, Fault Testing and Data Logging

## Disturbances and Fault

- Steady state response
- Positive pole to ground
- Negative pole to ground
- Pole-Pole fault
- Step up
- Step down
- Single phase to ground at Rectifier and Inverter side
- Three-phase to ground at Rectifier and Inverter side

## Data collection and training

- Combine 7 inputs and 2 outputs in single .CSV file
- Total number of sample matrix 1,35,010 x 9
- Neural Net Fitting App from MATLAB for training splits **in 70% for training, 15% for Validation and 15% for testing**
- Replaced the PID block with ANN directly with the same inputs



# Performance Comparison

An

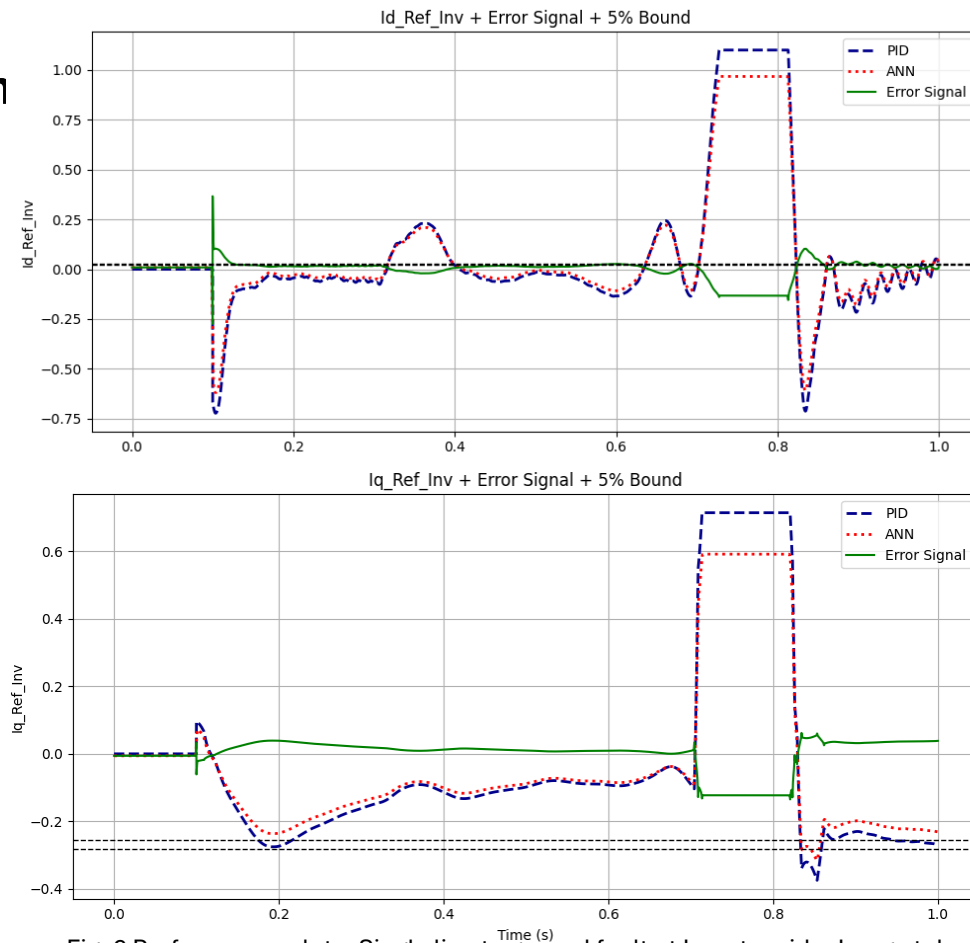


Fig. 6 Performance plots, Single line to ground fault at Inverter side, Image taken from MATLAB.

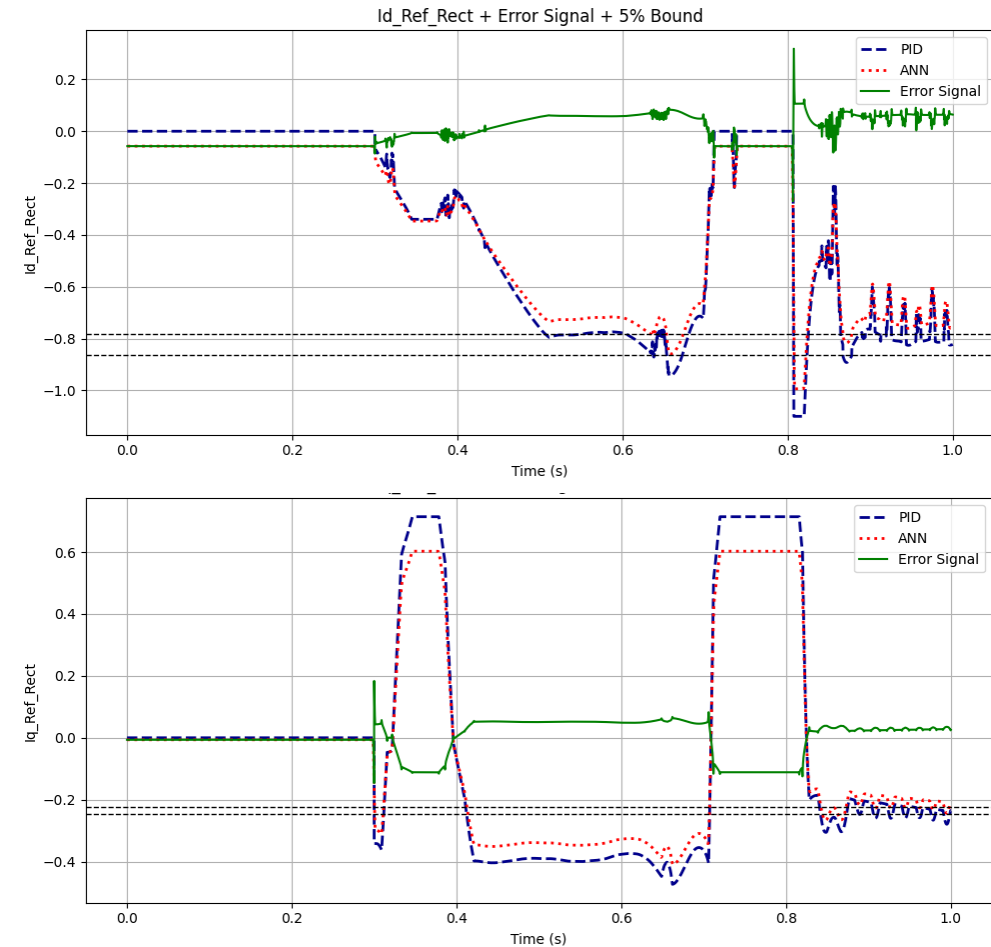


Fig. 7 Performance plots, Single line to ground fault at Rectifier side, Image taken from MATLAB.

# Conclusion & Future Work

## Conclusion

ANN outperform PID in terms of:

- Fault resilience
- Transient recovery
- Adaptability to disturbances
- Validate the approach of training the ANN model

## Future work

- Extend to multi-terminal HVDC systems
- Deploy in a real-time simulation environment

# References

1. B. Lu, Z. Zhang, Y. Ni, et al., “Artificial Neural Network Based Controller for VSC-HVDC,” *IEEE Transactions on Power Delivery*, vol. 34, no. 1, pp. 132–140, 2023.
2. A. K. Sahoo and R. Mohanty, “ANN Based Fault Detection in HVDC Transmission Systems,” *International Journal of Electrical Power & Energy Systems*, vol. 152, 2023.
3. M. Rashid, *Power Electronics Handbook*, Academic Press, 4th Edition, 2017. [Chapter: HVDC Converters and Control Strategies]
4. S. Bose, “PID Control: Design and Tuning Methods,” *Control Engineering Practice*, vol. 95, 2022, pp. 104302.
5. MATLAB Simulink Documentation: *Neural Network Fitting App & Simscape Electrical Toolbox*, The MathWorks, Inc.

THANK YOU FOR YOUR ATTENTION!

**OPEN FOR Q&A**